KEDGE dart v35: engineering report

v35 is v34 with the drill pods turned round: they hinge at the FRONT (x 50) and fold BACK along the flanks; deployed, the drills rake back toward the tether like tent pegs and the tow pushes each pod onto its printed deploy stop. Everything else is v34 (righting leg, tail tie, nose putty, press-fit couplers, cells, boards, firmware pin map). All numbers below come from the exact print geometry and the pipeline (run_all.sh), the righting sim (sim_righting) and flight35.py.

1. Summary

Item	Result
Body	Ø42 x 152 mm; folded envelope x -0.0 to 152.0, y ±40.5, z -21.0 to 21.3 (nothing behind the tail, nothing above the hull but a 0.3 mm knuckle blister)
Mass, centre of mass	249.6 g with catalogue bought masses and 14.6 g nose putty; CG folded [75.57, -0.1, 0.39], anchored [73.74, -0.1, -3.83], leg out [73.64, -0.1, 0.29] (mm; z from the hull axis)
Collision sweep	CLEAN: righting leg 0 to 178.0° in 179 steps, pods 0 to 46.75° in 48 steps, leg x pod cross sweep, every sub-part, exact meshes
Hard stops (mm³ of overlap just past each end)	deploy_at_angle 0.0, deploy_horn_+2.5deg 5.24, fold_at_0 0.0, fold_lug_-3deg 5.57, kick_rest_-0.5deg 0.0, kick_rest_+2deg 58.54, leg_fold_at_0 0.0, leg_fold_-2deg 429.87, ok True
Drills	deploy 46.75° (solved: the motor end in the hub stays 0.5 above flat ground), raked back, 43.5 mm deep, the tips about 119 mm from the nose tip (under the tail)
Launcher fit	FITS over 1684 launcher states (base/cad/parts/launch.py, parked)
Wire room	left corridor worst fill [0.08, 136.0], right [0.07, 136.0] (bundle area over free area; under 0.6 is fine)
Printed parts	81.22 cm³, every printed part watertight and one body

2. Layout and why

• Pods: pivot x 50, z 9.5 (z kept: the launcher's stop springs pass the dart at z -1.1 outboard of |y| 25, so the hubs and the arm-head lugs must stay above them), drill axis |y| 31.15. Folded they point aft along the flanks (auger tips at x 151); deploy 46.75° swings the tips down and aft to a backward rake.
• Parts: the v34 arms, augers, motors, couplers and pod caps are reused unchanged, turned round and swapped over (the v34 right pod is the v35 left pod), so the left pod now has the left-hand auger and the swapped drill leads. Only the hull halves, the tray and the righting leg change.
• Deploy stop: carved in the pivot block (the swept arm + HORN_GAP), now forward of and below the pivot; the fold stop under the head. The soil pushes the pods toward deploy in the tow, so the stop, not the worm, holds them.
• Interior: leg cradle | disc 1 | deploy cradle with DRV#1 / DRV#2 standing beside it | disc 1b (the B- spring on its rear face) | cells in line on the belly | rear disc | sockets and capacitor on the tail key. Deck 1 starts behind disc 1b. The deploy worm box sits under the righting leg's bridge: a channel in the bridge's underside and a hump in the hull top make room.
• The right pod covers the XIAO's USB-C window folded and deployed: flash over USB before the pods are fitted, then over WiFi (OTA).

3. Load paths and the tow

• Anchored tow: line -> the two holes in the tail face (hull) -> hull -> pivot blocks -> the carved deploy stops and the coupler bores -> arm heads -> drills -> soil. The soil pushes each pod onto its stop; the self-locking worm is not loaded.
• Righting sim (sim_righting/v35_check, study sim with layout front, one deploy worm on a rigid axle, line tied in the tail face, hull {'mass': 0.1921, 'cgX': 74.9}): righted loam 80/80, sand 20/20, named landing poses 20/20, fixed arrival paths 25 to 57° 40/40; median 7.0 s, slowest 17.1 s. Tow: deploy worm 0 N m (v34 tail tie: 0.94), deploy stops 4.237 N m for both pods at the peak (about 2.4 N m a pod scaled to 55 N, about 280 N on each stop face 8.5 mm from the pivot).
• Holding in the sim: loam 46.7 to 49.5 N (median 49.0), sand 17.1 to 17.3 N (median 17.2) (v34 tail tie: 79 of 80 held 55 N in loam, sand 18.3 N). The anchors now sit near the tail, so the pull at the tail pitches the nose up about the tail edge (about 12° at 45 N), taking weight off the hull and pulling the drills up and back until the sim's soil limit (25 N a drill in loam, the same in every direction) is reached. Each drill is loaded about 95 % sideways and only about 4 N along its axis (outward); the sim gives no credit for that, so a soil pull test decides the real number.
• Drill loads: side load on the auger hub -> arm bushing; push -> arm shoulder; pull -> nut A on the motor's threaded shaft. Righting: leg -> two couplers -> knuckle-wall bores and skin plugs. Launch: pusher -> flat tail end walls -> hull -> radial screws -> tray -> cells on the rear plate.

4. Mass properties (catalogue bought masses; weigh and update)

Pose	Mass g	CG x, y, z mm
folded	249.6	[75.57, -0.1, 0.39]
deployed	249.6	[73.74, -0.1, -3.83]
leg_out	249.6	[73.64, -0.1, 0.29]

Printed 100.8 g at 4 walls / 40% infill; bought parts and extras 148.8 g (incl. 14.6 g nose putty). Per-part volume, centroid and inertia: mass_props.json.

5. Launcher (parked base) and flight

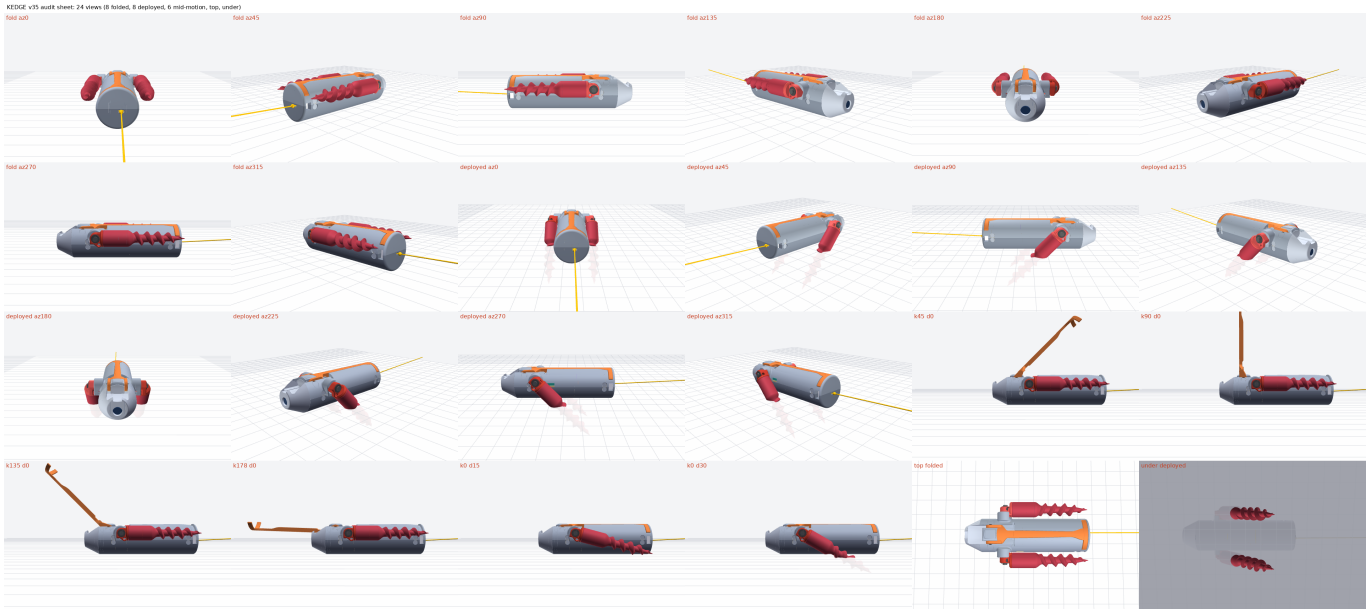
• Fit: FITS over 1684 states. Rest check: {'dart_vs_carriage_mm3': 0.0, 'dart_down_0p5_vs_carriage_mm3': 282.253}

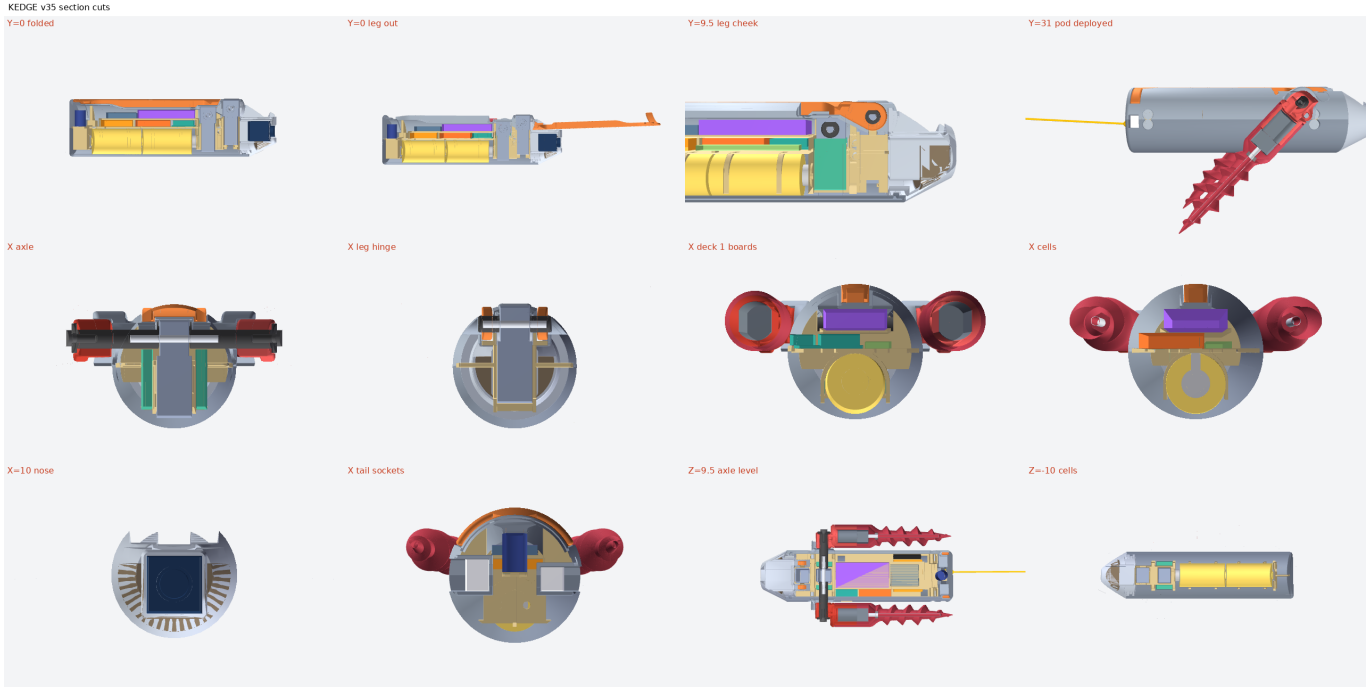
• Flight (flight35.py: flight34.py's strip model on the folded out35 geometry, line tied in the tail face, rails 6 to 47°): v35 nose first at every rail: True; tow arm 76.4 mm behind the CG, neutral speed 11.68 m/s (flight 6.5 to 6.9); tip-off: AoA max 23.3°, sideslip max 4.0°; 5° + 1 rad/s: AoA max 28.3°. v34 for comparison: nose first at every rail: True; tow arm 68.9 mm behind the CG, neutral speed 11.29 m/s (flight 6.5 to 6.9); tip-off: AoA max 28.0°, sideslip max 4.5°; 5° + 1 rad/s: AoA max 33.4°. The CG is about 7.6 mm further forward than v34, which more than pays for the pod hubs now at the front.
• What base needs: nothing new (the tail face, the tie holes and the envelope are v34's).

6. For dart-fw

• Pin map unchanged. The deploy turns the other way from v34 (pods hinge at the front): swap the deploy N20's two leads at DRV#1 AOUT1 / AOUT2 (or set the deploy sign to -1 in kedge_logic.h PINS); the deploy angle (46.75°) and its stops are as v34.
• Drills: DRILL_SIGN +1 unchanged; the swapped drill pair is now the LEFT pod's (the left-hand auger moved to the left).
• The righting sequence ran unchanged in the sim (pods-out roll on a flank, leg flip on the back): 100 % righted. USB-C is covered by the right pod once the pods are on: OTA for updates.

7. Pictures


KEDGE v35 audit sheet: 24 views (8 folded, 8 deployed, 6 mid-motion, top, under)
fold az0
fold az45
fold az90
fold az135
fold az180
fold az225
fold az270
fold az315
deployed az0
deployed az45
deployed az90
deployed az135
deployed az180
deployed az225
deployed az270
deployed az315
k45 d0
k90 d0
k135 d0
k178 d0
k0 d15
k0 d30
top folded
under deployed


KEDGE v35 section cuts
Y=0 folded
Y=0 leg out
Y=9.5 leg cheek
Y=31 pod deployed
X axle
X leg hinge
X deck 1 boards
X cells
X=10 nose
X tail sockets
Z=9.5 axle level
Z=-10 cells